# Foam engraving machine details

YD machinery equipment co.

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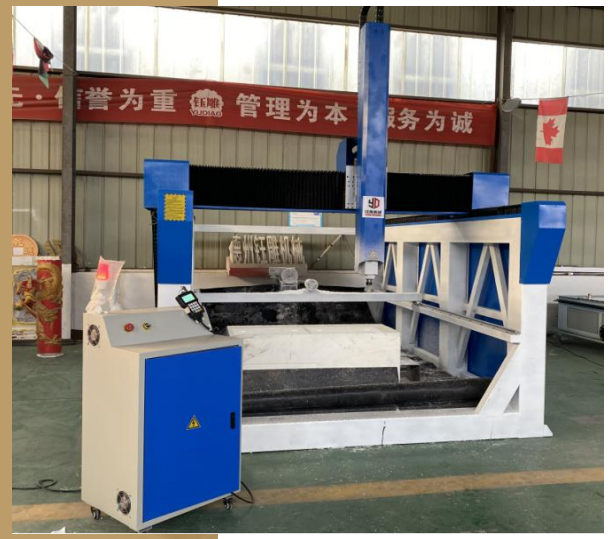
# 01

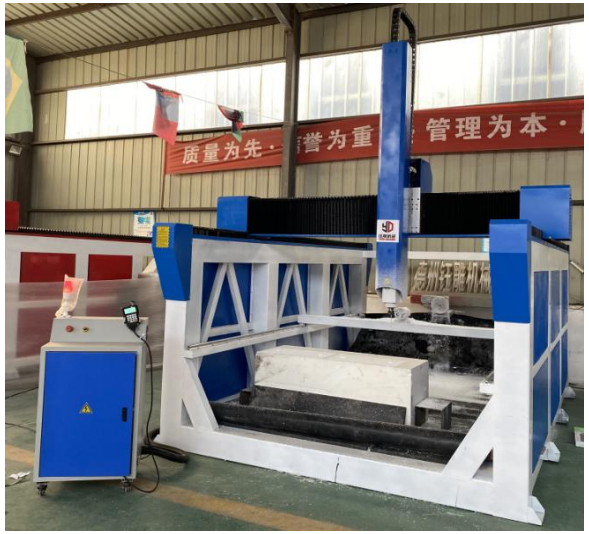
## Foam engraving machine working principle

# Machine overall picture



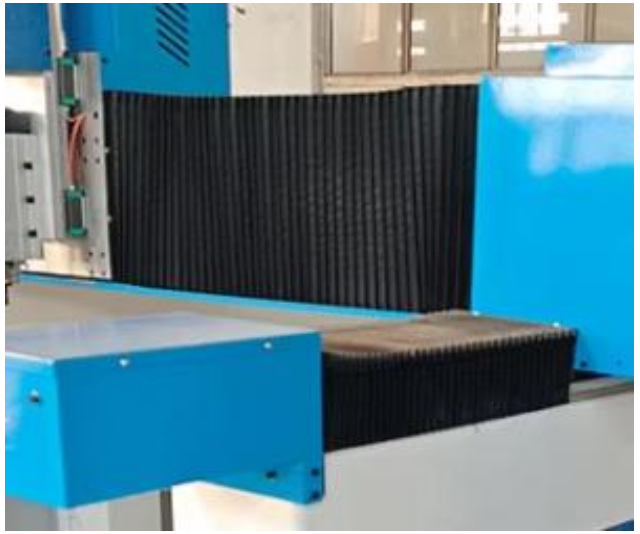

## Individual control cabinet

Avoid static interference, dubbed switch button and emergency stop button, to ensure the use of safety




## Model 2030

The machine adopts seamless welding one-piece thickened body, stable and durable, good quality, high quality.



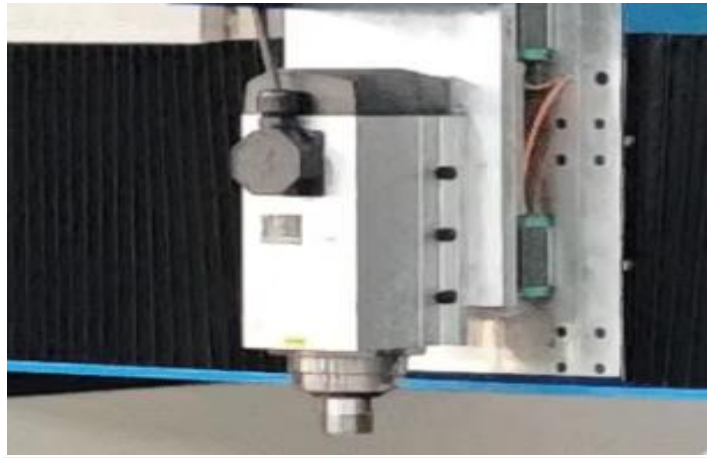
## Dustproofing of the whole machine

Effectively isolates dust and prolongs the life of the machine

# Composition of equipment

## Drive system

The drive system of a foam engraving machine usually includes stepper motors or servo motors (which can be customized) that are responsible for precisely controlling the movement of the engraving head.



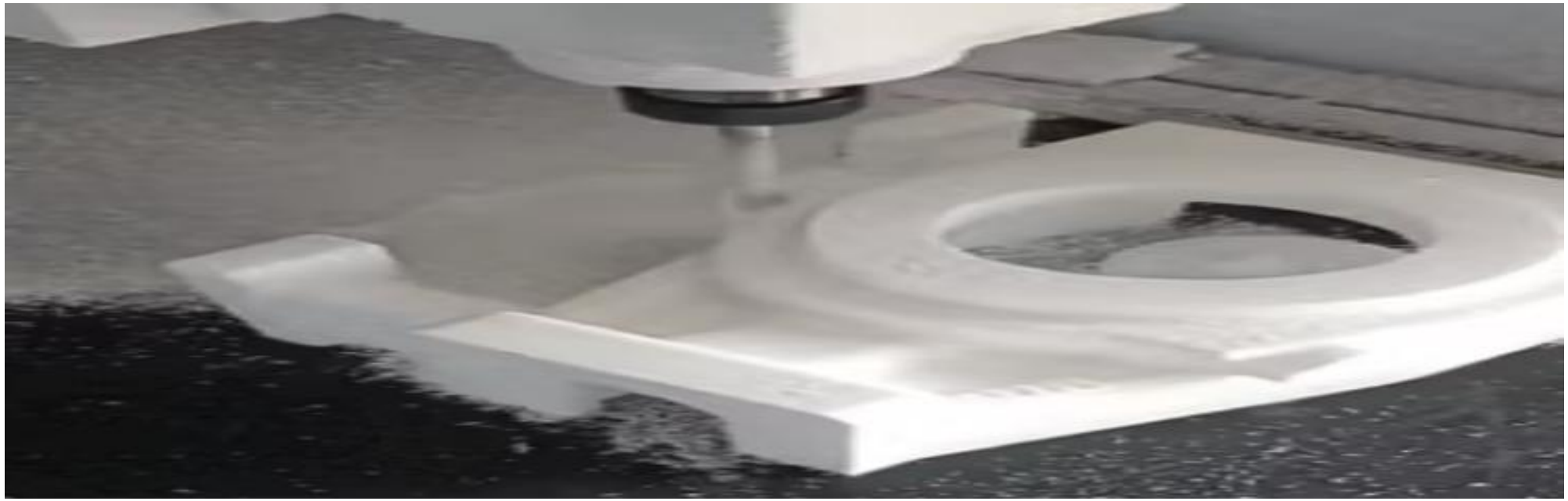
## Engraving head assemblies

The carving head assembly is the core component of the foam carving machine, which consists of the cutter and the rotating device for cutting the foam material.

## Heavier bed

The machine adopts seamless welding one-piece thickened body, stable and durable, bearing heavy, safer to use, more assured!

# Principle of engraving technology



## Mechanical Engraving Principles

Mechanical engraving utilizes a high-speed rotating tool to physically cut into the foam material to form the desired shape.

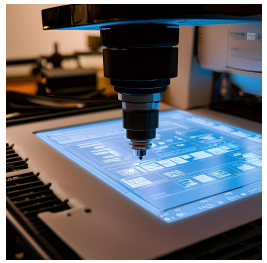
# Drive and Control Systems

## Stepper motor utilization

Stepper motors precisely control the movement of the engraving head to ensure an accurate engraving path

## Precise control of servo motors

Servo motors provide high-precision dynamic response to ensure speed and position control during the engraving process.



## Computer Numerical Controls

The computer numerical control system is programmed to control the starting, stopping and engraving process of the engraving machine to realize automatic operation.

# Explanation of engraving process

## Toolpath Programming

The engraving machine precisely cuts the foam material to form complex patterns by setting the movement path of the tool through the software.

## Foam material removal

During the carving process, the tool moves in a predetermined path, gradually removing excess foam and completing the carving.

# 02

# Foam carving machine application areas

# Advertising logo production

## Exhibition Design

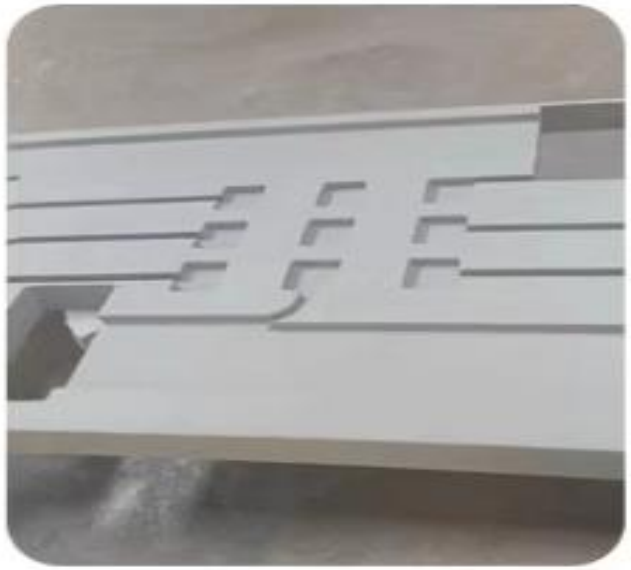
Foam carving machine is used in exhibition design to produce three-dimensional billboards and display models to enhance the visual effect.

## Outdoor signage production

Making outdoor billboards with foam carving machines is widely used in commercial advertising due to its light weight and easy installation.

## Customized store signs

Foam engraving machine can be customized personalized signboards for all kinds of stores to meet the signage needs of different businesses and enhance brand recognition.



# Art Deco Design

## Interior decorating

Foam carving machine can make a variety of interior decorations, such as relief wall decorations, adding a sense of space art.

## Holiday Festival Decoration

Create holiday decorations such as Halloween pumpkin lights and Santa Claus using a foam carving machine to create a festive atmosphere.

## Commercial Display Design

Foam engraving machine is widely used in commercial display, such as making product models and exhibition display boards to attract customers' attention.

## Theme Park Decoration

Foam carving machines can provide customized decorations for parks, such as cartoon characters and mythological scenes, to enhance the visitor experience.

# Model-making industry

## Architectural Modeling

Foam carving machines are widely used in architectural modeling to quickly and accurately carve out details of buildings.

## Anime Character Model

The carving machine can produce models of anime characters and provide fine three-dimensional images for anime exhibitions or collections.

# Special materials processing

## Architecture Modeling

Foam carving machine is widely used in architectural modeling, wood acrylic non-metal can also be carved, can quickly and accurately carve out the architectural details.

## Art Sculpture Creation

The artist utilizes a foam carving machine to create sculptures, creating works of art with varying forms and details.

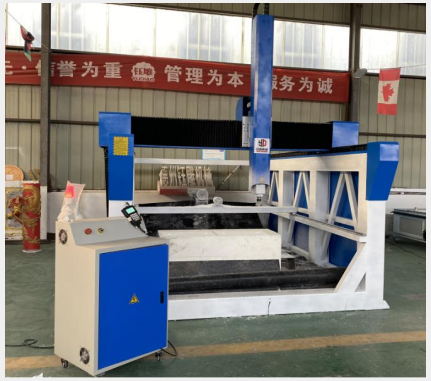
## Exhibition Design

In exhibition display design, foam carving machine can produce visually appealing display props and backdrops.

# 03

# Foam engraving machine technical parameters

# Equipment Size & Weight

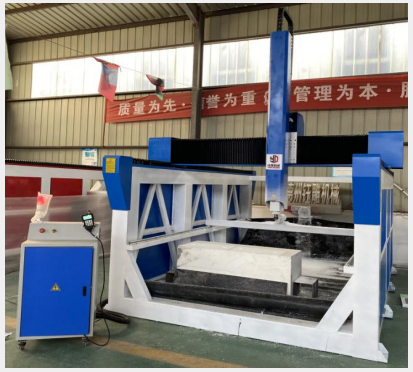



## Overall machine dimensions

Foam engraving machines typically measure 2000 x 3000 x 1500mm and are suitable for a wide range of working environments.

## Workbench area

The table area is approximately 2000x3000x1600mm, ensuring that there is plenty of room for large foam engraving.




## Equipment weight

The weight of the foam engraving machine is about 2 tons, which makes it easy to move and install in different places.

# Engraving accuracy and speed

## Engraving resolution

The engraving resolution of the foam engraving machine determines the fineness of detail, commonly ±0.1mm, etc..

## Engraving speed adjustment

Users can adjust the engraving speed according to the material and design requirements to achieve the best engraving effect and efficiency.

## Repeat Positioning Accuracy

Repeat positioning accuracy reflects the accuracy of the engraving machine when engraving the same pattern many times, is an important indicator of the performance of the machine.

# Engraving detailed parameters

| Porduct name | Styrofoam carving machine | Running motor | Stepper Hybrid Servo AC Pure Servo |
|---|---|---|---|
| Repeat Positioning Accuracy | ±0.01mm | | Fully automatic timed oiling system |
| Run command | G-code, F/S command compatible | operating system | Standard with Vivante/DSP |
| transmission system | X/Y gantry rack and pinion double drive | dust | Dustproofing of all aspects of the guide gears |
| Z-axis drive system | High Precision Ball Screws | stroke | 2000*3000*800mm |
| Spindle power | 6.0kw | control box | External Control Cabinet |
| Cylindrical engraving (optional) | 0-3000mm/10-50mm | Overall dimensions | 3000*4000*1600mm |
| Power Requirements | 380V/50Hz | processing speed | 20-30m/min |

# Compatible Material Types

## Types of Foam Materials

Foam carving machine can carve many kinds of foam, such as EPS, XPS and PU foam, to meet different production needs.

## Material size compatibility

The foam engraving machine supports a wide range of material sizes, with a maximum engravable size of 2000mm x 3000mm.

## Hardness of engraving material

The machine is capable of adapting to different hardnesses of foam materials, from soft to hard foam, ensuring engraving accuracy.

## Handling of special materials

In addition to conventional foams, the machine can also handle foams with special additives to enhance their durability.

# Software Support and Compatibility

## Supported file formats

The foam engraving machine is compatible with many file formats, such as STL, DXF, to ensure the smooth import and engraving of design files.

## Operating System Compatibility

The engraver software supports mainstream operating systems, including Windows and Mac OS, which is convenient for different user groups.

## Third-party software integration

The engraver software can be integrated with common CAD/CAM design software, such as AutoCAD and SolidWorks, to improve design efficiency.

# 04

## How to operate the foam engraving machine

# Equipment installation and commissioning

## Preparation of the installation environment

Choose a dry, well-ventilated space to install the foam carving machine to ensure stable operation of the equipment.

## Equipment hardware assembly

Follow the steps in the instruction manual to properly assemble the parts of the engraving machine, including the motor and guide rails.

## Software system configuration

Install the special software for the engraving machine and set the parameters to ensure the compatibility between the software and the hardware.

## Accuracy Calibration and Testing

Performs accuracy calibration of engraving machines and tests machine performance to ensure engraving accuracy and efficiency.

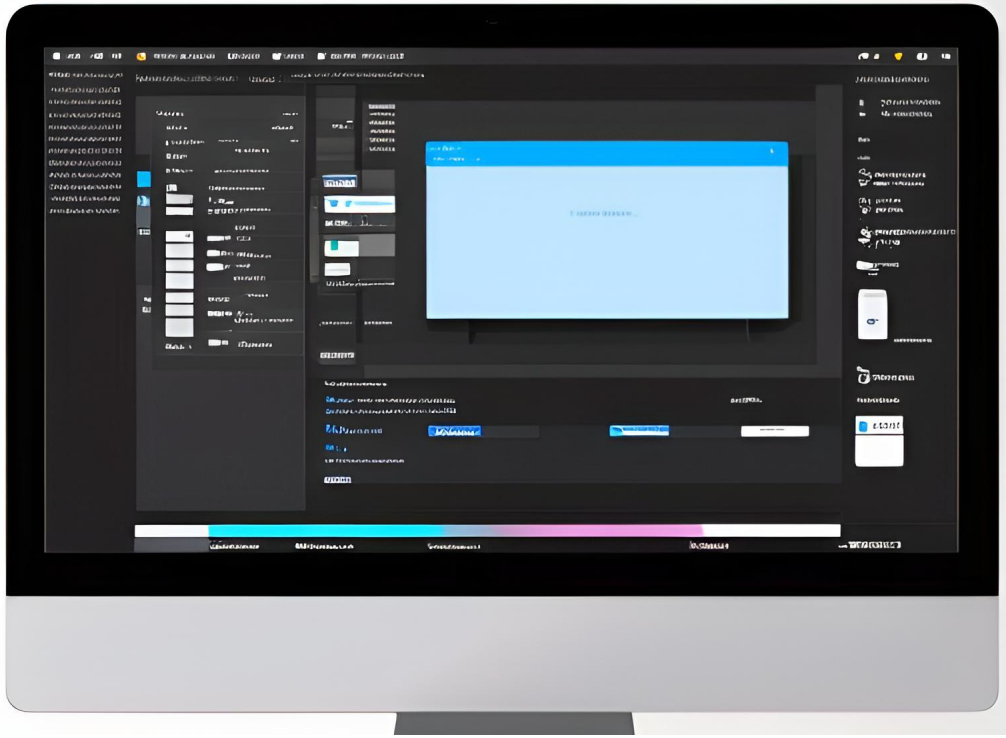
# Introduction to the software interface

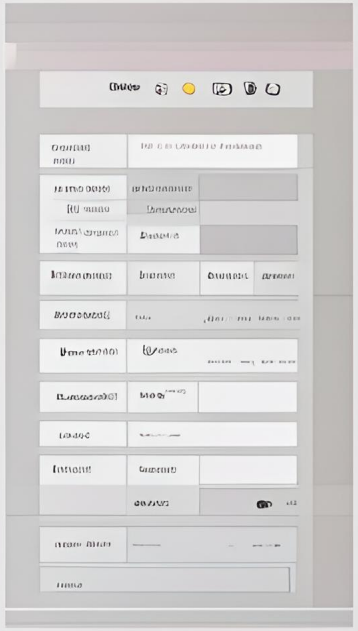
## Layout Overview

Introduces the main window layout of the software interface, including basic components such as toolbars, design areas, and preview windows.

## Function Shortcuts

List shortcuts to commonly used functions in the software, such as save, undo, copy, etc., for quick and easy operation.





# Engraving process steps

## Preparing materials for carving

Choose the right foam material to ensure that it is flat and undamaged, suitable for fine processing by the engraving machine

## Setting the engraving parameters

According to the design drawings and engraving requirements, set the appropriate speed, depth and tool path on the engraving machine control panel.

## Execution of engraving operations

Starts the engraving machine, monitors the engraving process to ensure stable machine operation and makes timely adjustments to avoid accumulation of errors.

# Frequently Asked Questions and Solutions

## Engraving machine won't start

Check that the power connections are correct, make sure the power switch is on, and check the fuse if necessary.

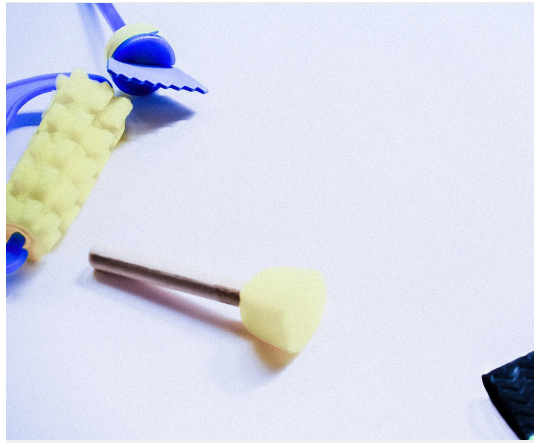
## Inaccurate engraving accuracy

Calibrate the positioning system of the engraving machine to ensure that the software settings are consistent with the actual engraving dimensions, and check the tool for wear and tear。

# 05

## Foam engraving machine maintenance

# Daily cleaning and maintenance

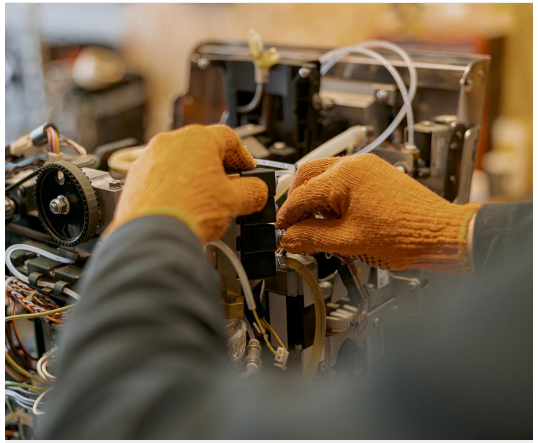


## Cleaning knives

Regularly clean the knives of your foam engraving machine with a special cleaner to maintain cutting accuracy and prolong service life.

## Lubricated guide rails

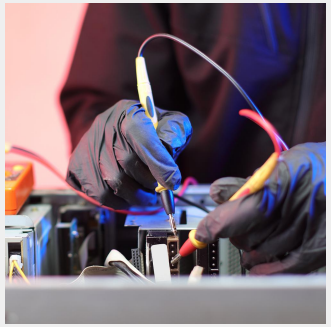
Regularly check and lubricate the guide rails of the engraving machine to ensure smooth operation and reduce wear and tear.



## Check circuit connections

Regularly check the circuit connections of the foam engraving machine to ensure that there is no looseness or damage, to ensure the safe and stable operation of the machine.

# Troubleshooting and Repair

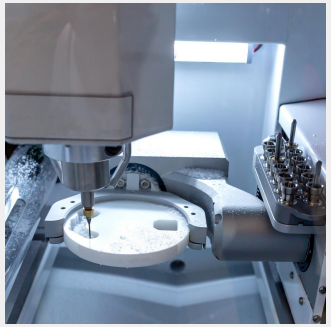


## Regular inspection of the electrical system

Make sure there are no loose circuit connections and check wires for frayed or shorted wires to prevent electrical failure.

## Cleaning and lubricating guide rails

Regularly clean the guide rails to remove dirt and debris to ensure that the engraving machine runs smoothly and reduces wear and tear.



## Replacement of worn tools

Regularly check the tool wear and replace it in time to ensure the engraving quality and machine precision.

# Parts replacement cycle

## Tool change intervals

Foam engraving machine tool according to the frequency of use and material hardness, usually every carving 100-200 square meters need to be replaced.

## Bearing replacement intervals

Bearings may wear out after long periods of operation and are generally recommended to be inspected and replaced after every 1,000 hours of use.

## Maintenance intervals for guide rails

The guide rail should be checked regularly, if you find wear and tear or strange noise, need to be replaced in time to ensure the precision of engraving

## Maintenance intervals for motors

The motor is the core component of the engraving machine and should be professionally inspected every six months and the carbon brushes should be replaced or repaired if necessary.

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